

Unit 1 Practice Problems

Foundations of Geometry Practice

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Unit 1 Practice Problems

Directions

Work each problem without looking at the answer bank. These Unit 1 problems include an answer bank only. Detailed worked solutions begin in Unit 3 as requested.

Problems

1. Explain what $\overleftrightarrow{AB}$ represents.
2. If A, B, C are collinear and B is between A and C , name the two opposite rays with endpoint B .
3. If B is between A and C , $AB = 3x + 2$, $BC = 2x + 7$, and $AC = 34$, find x , AB , and BC .
4. Point M is the midpoint of $\overline{AB}$. If $AM = 4x - 1$ and $MB = 2x + 9$, find x and AM .
5. If $m\angle ABD = 3y + 5$, $m\angle DBC = 2y + 17$, and $m\angle ABC = 72^\circ$, find y , $m\angle ABD$, and $m\angle DBC$.
6. Two angles form a linear pair. Their measures are $4x + 8$ and $6x - 18$. Find x and both angle measures.
7. Vertical angles measure $7x - 12$ and $4x + 33$. Find x and the angle measure.
8. Complementary angles measure $2x + 7$ and $x + 8$. Find x and both measures.
9. An angle measures $3x$, and its supplement measures $5x + 20$. Find x and the two angle measures.
10. Find the distance between $(-2, 5)$ and $(4, -3)$.
11. Find the midpoint of the segment with endpoints $(7, -1)$ and $(-3, 9)$.
12. Find the slope of the line through $(2, -5)$ and $(6, 7)$.
13. Are lines with slopes $\frac{2}{3}$ and $-\frac{3}{2}$ parallel, perpendicular, or neither?
14. Write the equation of the line with slope -2 through $(3, 5)$.
15. Write the equation of the line parallel to $y = 3x - 4$ through $(-1, 2)$.
16. Write the equation of the line perpendicular to $y = -\frac{1}{2}x + 7$ through $(4, -3)$.
17. Decide whether $2x - 4y = 8$ and $y = \frac{1}{2}x + 3$ are parallel, perpendicular, or neither.
18. For $P(1, 2)$ and $Q(7, 10)$, find the midpoint and distance.
19. Describe the construction of a perpendicular bisector of $\overline{AB}$.
20. Point P lies on the perpendicular bisector of $\overline{AB}$. If $PA = 3x + 2$ and $PB = 5x - 8$, find x and PA .
21. Classify $\triangle ABC$ with $A(0, 0)$, $B(4, 0)$, and $C(4, 3)$ as right, acute, or obtuse.
22. Classify quadrilateral $ABCD$ with $A(0, 0)$, $B(4, 1)$, $C(5, 5)$, $D(1, 4)$ as specifically as possible.
23. Explain the difference between a segment and a line.
24. When copying an angle, which tool transfers equal distances?
25. State the slope criteria for parallel and perpendicular nonvertical lines.

Answer Bank

1. The line through A and B , extending forever in both directions.
2. $\overrightarrow{BA}$ and $\overrightarrow{BC}$.
3. $x = 5$, $AB = 17$, $BC = 17$.
4. $x = 5$, $AM = 19$.
5. $y = 10$, $m\angle ABD = 35^\circ$, $m\angle DBC = 37^\circ$.
6. $x = 19$; angles 84° and 96° .
7. $x = 15$; each vertical angle is 93° .
8. $x = 25$; angles 57° and 33° .
9. $x = 20$; angles 60° and 120° .
10. 10.
11. $(2, 4)$.
12. 3.
13. Perpendicular.
14. $y = -2x + 11$.
15. $y = 3x + 5$.
16. $y = 2x - 11$.
17. Parallel.
18. Midpoint $(4, 6)$, distance 10.
19. Draw equal-radius arcs from A and B above and below the segment; connect the arc intersections.
20. $x = 5$, $PA = 17$.
21. Right triangle.
22. Rhombus, not a square.
23. A segment has two endpoints and finite length; a line extends forever with no length.
24. Compass.
25. Parallel: equal slopes. Perpendicular: negative reciprocal slopes, product -1 .